

Digital Circuit Homework

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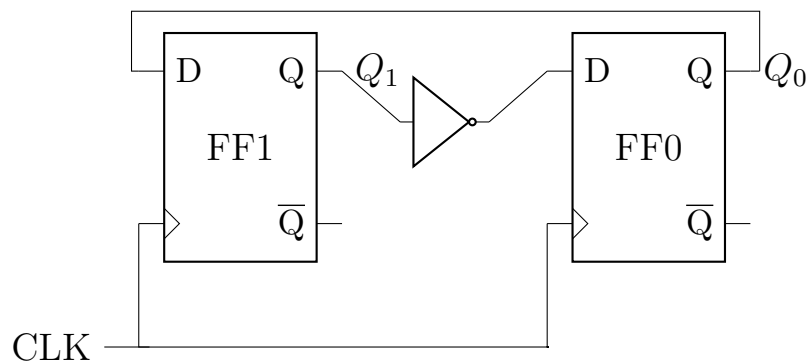
Deadline: 23:59 June 15, 2026

Instructions

- Please write your answers clearly and show the main steps of your reasoning.
- Except for the screenshots of HDLBits programming problem, **please submit handwritten solutions.**
- For the HDLBits problem, submit screenshots showing successful completion. Include your code if it is not clearly visible in the screenshot.
- Late submissions may result in a score deduction.
- Total: **100 pts.**

Problem 1.**[15 pts]**

Analyze the synchronous sequential circuit shown below. The circuit contains two positive-edge-triggered D flip-flops. The outputs of the flip-flops are denoted by Q_1 and Q_0 .



(a) Derive the Boolean expressions for the D inputs D_1 and D_0 . **[5 pts]**

(b) Write the complete state transition table. Use Q_1Q_0 as the present state and $Q_1^+Q_0^+$ as the next state. **[5 pts]**

(c) Draw the state transition diagram and determine whether the generated state sequence has the Gray-code property. Briefly justify your answer.

[5 pts]

Problem 2.**[15 pts]**

Build a circuit that realizes the next-state behavior of a positive-edge-triggered **JK flip-flop** using one positive-edge-triggered **D flip-flop** and basic logic gates. Ignore propagation delay, setup time, and hold time.

- (1) Complete the JK flip-flop next-state table below.

[5 pts]

J	K	Current State Q	Next State Q_{next}
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

- (2) Derive the simplified Boolean equation for the D input in terms of J , K , and the present state Q . That is, find $D = f(J, K, Q)$.

[5 pts]

$D =$ _____

- (3) Draw the circuit implementation using inputs J and K , one D flip-flop, and NOT, AND, and OR gates.

[5 pts]

Problem 3.**[15 pts]**

You are provided with several $32\text{K} \times 8\text{-bit}$ SRAM chips. Design a memory system with total capacity

$$64\text{K} \times 16 \text{ bits.}$$

Assume the memory is organized as 64K addressable locations, and each addressable location stores one 16-bit data word. The CPU has a 16-bit address bus ($A_{15}, A_{14}, \dots, A_0$), where each address selects one 16-bit memory location. The CPU also has a 16-bit data bus ($D_{15}, D_{14}, \dots, D_0$).

Each SRAM chip has an active-low chip-select input $\overline{CS}$. You only need to show the address bus, data bus, and chip-select logic; read/write control signals are omitted.

- (a) Determine the total number of $32\text{K} \times 8\text{-bit}$ SRAM chips required. Briefly explain how width expansion and depth expansion are achieved. **[5 pts]**

- (b) Draw a block-level circuit diagram showing the connections of the address bus, data bus, and active-low chip-select signals ($\overline{CS}$). Use the most significant address bit A_{15} and one NOT gate for address decoding. **[10 pts]**

Problem 4.**[20 pts]**

A 10-bit Successive Approximation Register (SAR) ADC has an input range $0 \leq V_{in} < 5 \text{ V}$ and operates at a clock frequency of 1 MHz.

Assume an ideal straight-binary ADC with

$$\Delta = \frac{V_{ref}}{2^N},$$

where $V_{ref} = 5 \text{ V}$ and Δ is the voltage represented by one least significant bit, LSB.

Assume each SAR bit decision takes exactly one clock cycle. Ignore sampling time, DAC settling time, comparator delay, and all other overhead.

Also assume the output code is obtained by truncation:

$$N_{out} = \left\lfloor \frac{V_{in}}{\Delta} \right\rfloor.$$

For reconstruction, use $V_{rec} = N_{out}\Delta$.

(a) Calculate the minimum conversion time required for one sample, in μs . **[5 pts]**

(b) If the analog input voltage is $V_{in} = 3.2 \text{ V}$, determine the ADC output code in decimal. **[5 pts]**

(c) Determine the reconstructed voltage corresponding to the digital output code found in part (b), in volts. **[5 pts]**

(d) Calculate the quantization error, in volts. Define the quantization error as

$$e_q = V_{in} - V_{rec}.$$

[5 pts]

Problem 5.**[15 pts]**

Consider a synchronous sequential circuit consisting of two D flip-flops, FF1 and FF2, with a combinational logic block between them. The circuit uses a single common clock.

The timing parameters are:

- Clock frequency: $f_{clk} = 50$ MHz
- Flip-flop maximum clock-to-Q propagation delay: $t_{pcq} = 2.5$ ns
- Flip-flop minimum clock-to-Q contamination delay: $t_{ccq} = 1.5$ ns
- Flip-flop setup time: $t_{setup} = 3.0$ ns
- Flip-flop hold time: $t_{hold} = 2.0$ ns

(a) Calculate the clock period T_{clk} . **[5 pts]**

(b) Determine the maximum allowable propagation delay of the combinational logic block, $t_{pd,max}$, so that there is no setup time violation. **[5 pts]**

(c) Determine the minimum required contamination delay of the combinational logic block, $t_{cd,min}$, so that there is no hold time violation. **[5 pts]**

Problem 6.**[20 pts]**

Complete the following three sequential logic exercises on HDLBits. For each exercise, submit a screenshot showing successful completion and your code. The original URLs are provided below.

(1) Multiplexer and D Flip-Flop (Mt2015 muxdff)**[7 pts]**Link: https://hdlbits.01xz.net/wiki/Mt2015_muxdff**(2) Simple FSM 1: Synchronous Reset (Fsm1s)****[7 pts]**Link: <https://hdlbits.01xz.net/wiki/Fsm1s>**(3) Testbench for T Flip-Flop (Tb/tff)****[6 pts]**Link: <https://hdlbits.01xz.net/wiki/Tb/tff>